ever since their birth. Once in a great while one of them reaches its critical, smallest size and explodes.[39] It was only by chance—an extremely improbable occurrence—that one exploded while hurtling past our ship, and it is exceedingly unlikely that our starship will ever encounter another such hole."

Relieved, you order your crew to begin repairs on the ship while you and your mates embark on your telescopic study of the 45-solar-mass, rapidly spinning hole below you.

The hole's spin is obvious not only from the swirl of infalling atoms, but also from the shape of the bright-ringed black spot it makes on the sky below you: The black spot is squashed, like a pumpkin; it bulges at its equator and is flattened at its poles. The centrifugal force of the hole's spin, pushing outward, creates the bulge and flattening.[40] But the bulge is not symmetric; it looks larger on the right edge of the disk, which is moving away from you as the horizon spins, than on the left edge. DAWN explains that this is because the horizon can capture rays of starlight more easily if they move toward you along its right edge, against the direction of its spin, than along its left edge, with its spin.

By measuring the shape of the spot and comparing it with general relativity's black-hole formulas, Bret infers that the hole's spin angular momentum is 96 percent of the maximum allowed for a hole of its mass. And from this angular momentum and the hole's mass of 45 Suns you compute other properties of the hole, including the spin rate of its horizon, 270 revolutions per second, and its equatorial circumference, 533 kilometers.

The spin of the hole intrigues you. Never before could you observe a spinning hole up close. So with pangs of conscience you ask for and get a volunteer robot, to explore the neighborhood of the horizon and transmit back his experiences. You give the robot, whose name is Kolob, careful instructions: "Descend to ten meters above the horizon and there blast your rockets to hold yourself at rest, hovering directly below the starship. Use your rockets to resist both the inward pull of gravity and the tornado-like swirl of space."